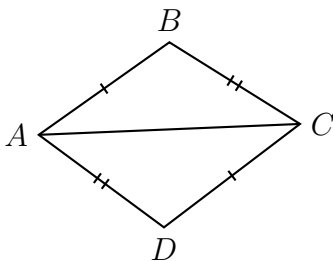


# Completing Two-Column Triangle-Congruence Proofs

*filling in reasons, filling in statements, and finishing with CPCTC*

- Every line needs a reason: a *given*, a definition, the reflexive property, a theorem (vertical angles, alternate interior angles), or a congruence criterion.
- A shared side or angle enters by the **reflexive property**; it is never marked on the figure.
- **CPCTC** — corresponding parts of congruent triangles are congruent — is what you use *after* the criterion, never before it.
- Diagrams show the marks and the labelling; all measurements are in the problem text.

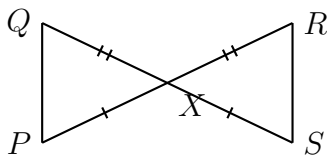
1. **Given:**  $\overline{AB} \cong \overline{CD}$  and  $\overline{BC} \cong \overline{DA}$ . **Prove:**  $\triangle ABC \cong \triangle CDA$ .



Fill in the reason column.

| Statements                             | Reasons |
|----------------------------------------|---------|
| 1. $\overline{AB} \cong \overline{CD}$ | 1.      |
| 2. $\overline{BC} \cong \overline{DA}$ | 2.      |
| 3. $\overline{AC} \cong \overline{CA}$ | 3.      |
| 4. $\triangle ABC \cong \triangle CDA$ | 4.      |

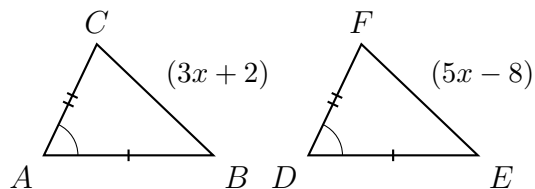
2.  $\overline{PR}$  and  $\overline{QS}$  intersect at  $X$ . **Given:**  $\overline{PX} \cong \overline{SX}$  and  $\overline{QX} \cong \overline{RX}$ . **Prove:**  $\triangle PQX \cong \triangle SRX$ .



Fill in the reason column.

| Statements                             | Reasons |
|----------------------------------------|---------|
| 1. $\overline{PX} \cong \overline{SX}$ | 1.      |
| 2. $\angle PXQ \cong \angle SXR$       | 2.      |
| 3. $\overline{QX} \cong \overline{RX}$ | 3.      |
| 4. $\triangle PQX \cong \triangle SRX$ | 4.      |

**3. Given:**  $\overline{AB} \cong \overline{DE}$ ,  $\angle A \cong \angle D$ ,  $\overline{AC} \cong \overline{DF}$ , so  $\triangle ABC \cong \triangle DEF$ . In centimetres,  $BC = 3x + 2$  and  $EF = 5x - 8$ .



(a) Name the criterion the three givens establish, and name the reason that then entitles you to write  $\overline{BC} \cong \overline{EF}$ .

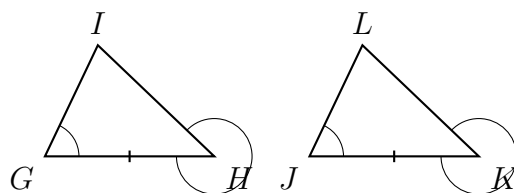
(b) Find  $x$ .

Answer: \_\_\_\_\_

(c) Find  $BC$ .

Answer: \_\_\_\_\_ cm

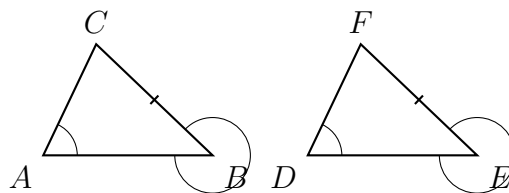
4. **Given:**  $\angle G \cong \angle J$ ,  $\overline{GH} \cong \overline{JK}$ ,  $\angle H \cong \angle K$ . **Prove:**  $\triangle GHI \cong \triangle JKL$ .



This time the reasons are printed. Fill in the statement column.

| Statements | Reasons  |
|------------|----------|
| 1.         | 1. Given |
| 2.         | 2. Given |
| 3.         | 3. Given |
| 4.         | 4. ASA   |

**5. Given:**  $\angle A \cong \angle D$ ,  $\angle B \cong \angle E$ ,  $\overline{BC} \cong \overline{EF}$ , so  $\triangle ABC \cong \triangle DEF$ . In degrees,  $m\angle C = 4x - 9$  and  $m\angle F = 2x + 15$ .



(a) The marked side is not between the two marked angles. Name the criterion, and explain why ASA does not apply here.

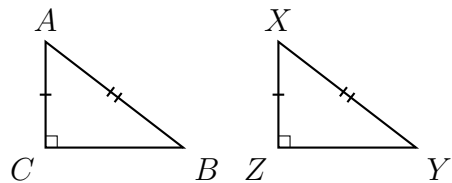
(b) Find  $x$ .

Answer: \_\_\_\_\_

(c) Find  $m\angle C$ .

Answer: \_\_\_\_\_ degrees

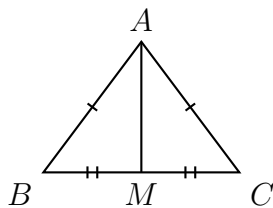
**6.**  $\triangle ABC$  and  $\triangle XYZ$  have right angles at  $C$  and  $Z$ . **Given:**  $\overline{AB} \cong \overline{XY}$  and  $\overline{CA} \cong \overline{ZX}$ . **Prove:**  $\triangle ABC \cong \triangle XYZ$ .



Fill in the reason column.

| Statements                                                 | Reasons |
|------------------------------------------------------------|---------|
| 1. $\angle C$ and $\angle Z$ are right angles              | 1.      |
| 2. $\triangle ABC$ and $\triangle XYZ$ are right triangles | 2.      |
| 3. $\overline{AB} \cong \overline{XY}$                     | 3.      |
| 4. $\overline{CA} \cong \overline{ZX}$                     | 4.      |
| 5. $\triangle ABC \cong \triangle XYZ$                     | 5.      |

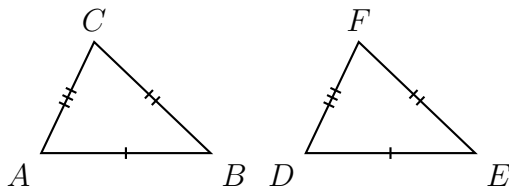
**7.**  $M$  is the midpoint of  $\overline{BC}$ . **Given:**  $\overline{AB} \cong \overline{AC}$  and  $M$  is the midpoint of  $\overline{BC}$ .  
**Prove:**  $\triangle ABM \cong \triangle ACM$ .



The reasons are printed. Fill in the statement column.

| Statements | Reasons                             |
|------------|-------------------------------------|
| 1.         | 1. Given                            |
| 2.         | 2. Given                            |
| 3.         | 3. Definition of midpoint           |
| 4.         | 4. Reflexive property of congruence |
| 5.         | 5. SSS                              |

**8. Given:**  $\overline{AB} \cong \overline{DE}$ ,  $\overline{BC} \cong \overline{EF}$ ,  $\overline{CA} \cong \overline{FD}$ , so  $\triangle ABC \cong \triangle DEF$ . In degrees,  $m\angle B = 5x + 4$  and  $m\angle E = 3x + 26$ .



(a) Name the criterion, and name the reason that lets you conclude  $\angle B \cong \angle E$  once the triangles are congruent.

(b) Find  $x$ .

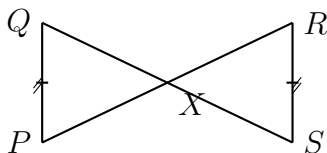
Answer: \_\_\_\_\_

(c) Find  $m\angle B$ .

Answer: \_\_\_\_\_ degrees



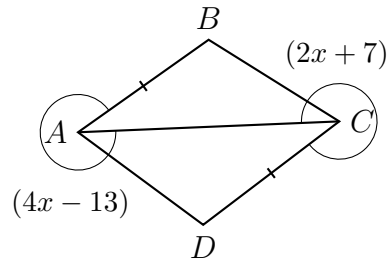
**9.** In quadrilateral  $PQRS$  the segments  $\overline{PQ}$  and  $\overline{SR}$  are parallel, and the diagonals meet at  $X$ . **Given:**  $\overline{PQ} \parallel \overline{SR}$  and  $\overline{PQ} \cong \overline{SR}$ . **Prove:**  $\triangle PQX \cong \triangle SRX$ .



Fill in the reason column.

| Statements                                 | Reasons |
|--------------------------------------------|---------|
| 1. $\overline{PQ} \parallel \overline{SR}$ | 1.      |
| 2. $\angle QPX \cong \angle RSX$           | 2.      |
| 3. $\angle PXQ \cong \angle SXR$           | 3.      |
| 4. $\overline{PQ} \cong \overline{SR}$     | 4.      |
| 5. $\triangle PQX \cong \triangle SRX$     | 5.      |

**10.** In quadrilateral  $ABCD$  the diagonal  $\overline{AC}$  is drawn. **Given:**  $\overline{AB} \cong \overline{CD}$  and  $\angle BAC \cong \angle DCA$ . In centimetres,  $BC = 2x + 7$  and  $DA = 4x - 13$ .



(a) Write the whole two-column proof that  $\triangle BAC \cong \triangle DCA$ . Use four numbered lines, each with its reason.

(b) Find  $x$ .

Answer: \_\_\_\_\_

(c) Find  $BC$ .

Answer: \_\_\_\_\_ cm